WORKED SOLUTION

STEP III 2020, Question 2

Pure • Hyperbolic Curve Sketching

STEP III 2020, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Differentiate implicitly, and keep using $\cosh^2 = 1 + \sinh^2$ to turn everything back into sines: the second-derivative condition then factorises, with one factor being the curve's own equation. For the line $x + y = a$, substitute $y = a - x$ and write everything in exponentials; a real point of intersection needs a non-negative discriminant, and that gives the bound on $\cosh a$.

Solution

Part (i)

Since sinh is a bijection of $\mathbb{R}$, for each x there is exactly one y with $\sinh y = 2k - \sinh x$. So C is the graph of a function $y = \operatorname{arsinh}(2k - \sinh x)$, defined for all real x, and we may differentiate implicitly:

$$\cosh x + \cosh y \,\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\cosh x}{\cosh y}.$$

Since $\cosh x \geq 1$, the gradient is never zero (it is always negative), so C has no stationary points.

Differentiating again, writing $y' = \mathrm{d}y/\mathrm{d}x$,

$$\sinh x + \sinh y\,(y')^2 + \cosh y\, y'' = 0.$$

Substitute $y' = -\cosh x / \cosh y$ and multiply through by $\cosh^2 y$:

$$\begin{aligned}
\cosh^3 y\, y'' &= -\left(\sinh x \cosh^2 y + \sinh y \cosh^2 x\right) \\
&= -\left(\sinh x\,(1 + \sinh^2 y) + \sinh y\,(1 + \sinh^2 x)\right) \\
&= -(\sinh x + \sinh y) - \sinh x \sinh y\,(\sinh y + \sinh x) \\
&= -(\sinh x + \sinh y)(1 + \sinh x \sinh y).
\end{aligned}$$

On C the first factor is $2k$, so

$$y'' = -\frac{2k\,(1+\sinh x \sinh y)}{\cosh^3 y}.$$

Since $k > 0$ and $\cosh y > 0$, this gives $y'' = 0$ if and only if $1 + \sinh x \sinh y = 0$, as required.

Ask yourself Is $y'' = 0$ enough to give a point of inflection?

No: we also need y'' to change sign. Put $s = \sinh x$ and $t = \sinh y$. At a point with $y'' = 0$ we have $s + t = 2k$ and $st = -1$, so s and t are the roots of $u^2 - 2ku - 1 = 0$:

$$u = k \pm \sqrt{k^2+1}.$$

One root is positive and one negative, and s can take either value, with t the other. Along the curve, $1 + st = 1 + s(2k - s) = -(s^2 - 2ks - 1)$ is a quadratic in s with two simple roots, so it changes sign at each one. Since $s = \sinh x$ is increasing in x, y'' does change sign at both points. The points of inflection are therefore

$$\left(\operatorname{arsinh}(k+\sqrt{k^2+1}),\ \operatorname{arsinh}(k-\sqrt{k^2+1})\right) \quad \text{and}$$
$$\left(\operatorname{arsinh}(k-\sqrt{k^2+1}),\ \operatorname{arsinh}(k+\sqrt{k^2+1})\right).$$

They are reflections of each other in $y = x$, which we expect, since the equation of C is symmetric in x and y.

Part (ii)

On the line, $y = a - x$. Writing $2\sinh u = \mathrm{e}^u - \mathrm{e}^{-u}$, the equation of C doubled becomes

$$\mathrm{e}^x - \mathrm{e}^{-x} + \mathrm{e}^{a-x} - \mathrm{e}^{x-a} = 4k.$$

Multiplying by e^x gives $\mathrm{e}^{2x} - 1 + \mathrm{e}^a - \mathrm{e}^{2x-a} = 4k\mathrm{e}^x$, that is,

$$\mathrm{e}^{2x}(1 - \mathrm{e}^{-a}) - 4k\mathrm{e}^x + (\mathrm{e}^a - 1) = 0,$$

which is the required equation.

Ask yourself What does the existence of such a point tell us about this equation?

It is a quadratic in $X = \mathrm{e}^x$ (provided its leading coefficient is non-zero) and it must have a real root.

First, $a \neq 0$: if $a = 0$ the equation collapses to $-4k\mathrm{e}^x = 0$, which is impossible since $k > 0$. Hence $\cosh a > 1$.

With $a \neq 0$ the leading coefficient $1 - \mathrm{e}^{-a}$ is non-zero, so the quadratic has a real root only if its discriminant is non-negative:

$$16k^2 - 4(1 - \mathrm{e}^{-a})(\mathrm{e}^a - 1) \geq 0.$$

Now $(1-\mathrm{e}^{-a})(\mathrm{e}^{a}-1) = \mathrm{e}^{a}-2+\mathrm{e}^{-a} = 2\cosh a - 2$, so the condition is $16k^2 \geq 8(\cosh a - 1)$, i.e. $\cosh a \leq 2k^2+1$. Together,

$$1 < \cosh a \leq 2k^2+1.$$

So every point of C has $x+y$ between $-2\operatorname{arsinh} k$ and $2\operatorname{arsinh} k$ (because $\cosh(2\operatorname{arsinh} k) = 1+2\sinh^2(\operatorname{arsinh} k) = 1+2k^2$), and $x+y \neq 0$.

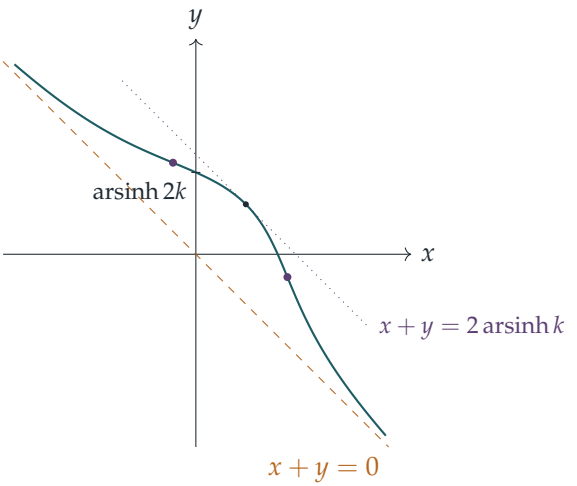
In fact $x+y > 0$ on C. If $a < 0$ then $1-\mathrm{e}^{-a} < 0$ and $\mathrm{e}^{a}-1 < 0$, so all three coefficients of the quadratic are negative and it has no positive root X. (Equivalently, $\sinh x + \sinh y = 2\sinh\frac{x+y}{2}\cosh\frac{x-y}{2}$, which has the sign of $x+y$.) The largest value $x+y = 2\operatorname{arsinh} k$ occurs at $x = y = \operatorname{arsinh} k$, where the discriminant vanishes and the line just touches C.

Part (iii)

Collecting what we know:

- C is symmetric in the line $y = x$ and crosses it only at $(\operatorname{arsinh} k, \operatorname{arsinh} k)$, where the gradient is -1.
- y is defined for all x and the gradient is always negative. The curve meets the axes at $(\operatorname{arsinh} 2k, 0)$ and $(0, \operatorname{arsinh} 2k)$.
- $0 < x+y \leq 2\operatorname{arsinh} k$, so C lies in the strip above the line $x+y=0$ and on or below the line $x+y = 2\operatorname{arsinh} k$, which touches C at its point on $y = x$.
- As $x \to +\infty$, $\sinh y = 2k - \sinh x \to -\infty$, and $\mathrm{e}^{-y} \approx \mathrm{e}^{x} - 4k$, so $x+y \approx -\ln(1-4k\mathrm{e}^{-x}) \approx 4k\mathrm{e}^{-x} \to 0^{+}$. By symmetry the same happens as $x \to -\infty$. So $x+y=0$ is an asymptote at both ends, approached from above.
- By the formula for y'', the curve is concave (below its tangents) between the two points of inflection, where $1+\sinh x \sinh y > 0$ (for example at the point on $y = x$), and convex outside them.

The sketch shows the case $k = 1$, with the points of inflection marked.



Checks

- With $k = 1$ the points of inflection are at $\left(\operatorname{arsinh}(1+\sqrt{2}), \operatorname{arsinh}(1-\sqrt{2})\right) \approx (1.615, -0.403)$ and its mirror image; checking second differences of $y = \operatorname{arsinh}(2 - \sinh x)$ on either side confirms y'' changes sign there, and a scan over $-8 \le x \le 8$ finds no other zeros of y''.
- The formula $y'' = -2k(1 + \sinh x \sinh y)/\cosh^3 y$ agrees with numerical second derivatives at many random points and for several values of k.
- For $k = 1$ and $x = 6$, $x + y \approx 0.00996$, close to the predicted $4\mathrm{e}^{-6} \approx 0.00992$: the asymptote is approached from above, as claimed.
- At $x = y = \operatorname{arsinh} k$, $\cosh(x+y) = 1 + 2\sinh^2 x = 1 + 2k^2$: the upper bound in (ii) is attained.

Common slips

- Stopping at $y'' = 0$ without checking that y'' changes sign. Here it does, but it needs saying.
- Forgetting to rule out $a = 0$, which is what makes the left-hand inequality strict.
- Only writing down one point of inflection. Either root of $u^2 - 2ku - 1 = 0$ can be $\sinh x$, giving two points mirrored in $y = x$.

Beyond STEP

The rotated coordinates $u = x + y$, $v = x - y$ make this curve transparent: the identity $\sinh x + \sinh y = 2\sinh\frac{u}{2}\cosh\frac{v}{2}$ turns C into $\sinh\frac{u}{2} = k\operatorname{sech}\frac{v}{2}$, a single hump over the asymptote $u = 0$. Choosing coordinates adapted to a symmetry is a theme throughout

first-year calculus and linear algebra (diagonalising a quadratic form is the same idea). Implicit curves, and when an equation $F(x, y) = 0$ really defines y as a smooth function of x, are the subject of the implicit function theorem; Bartle and Sherbert, *Introduction to Real Analysis*, gives a careful treatment of the one-variable analysis behind it.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.